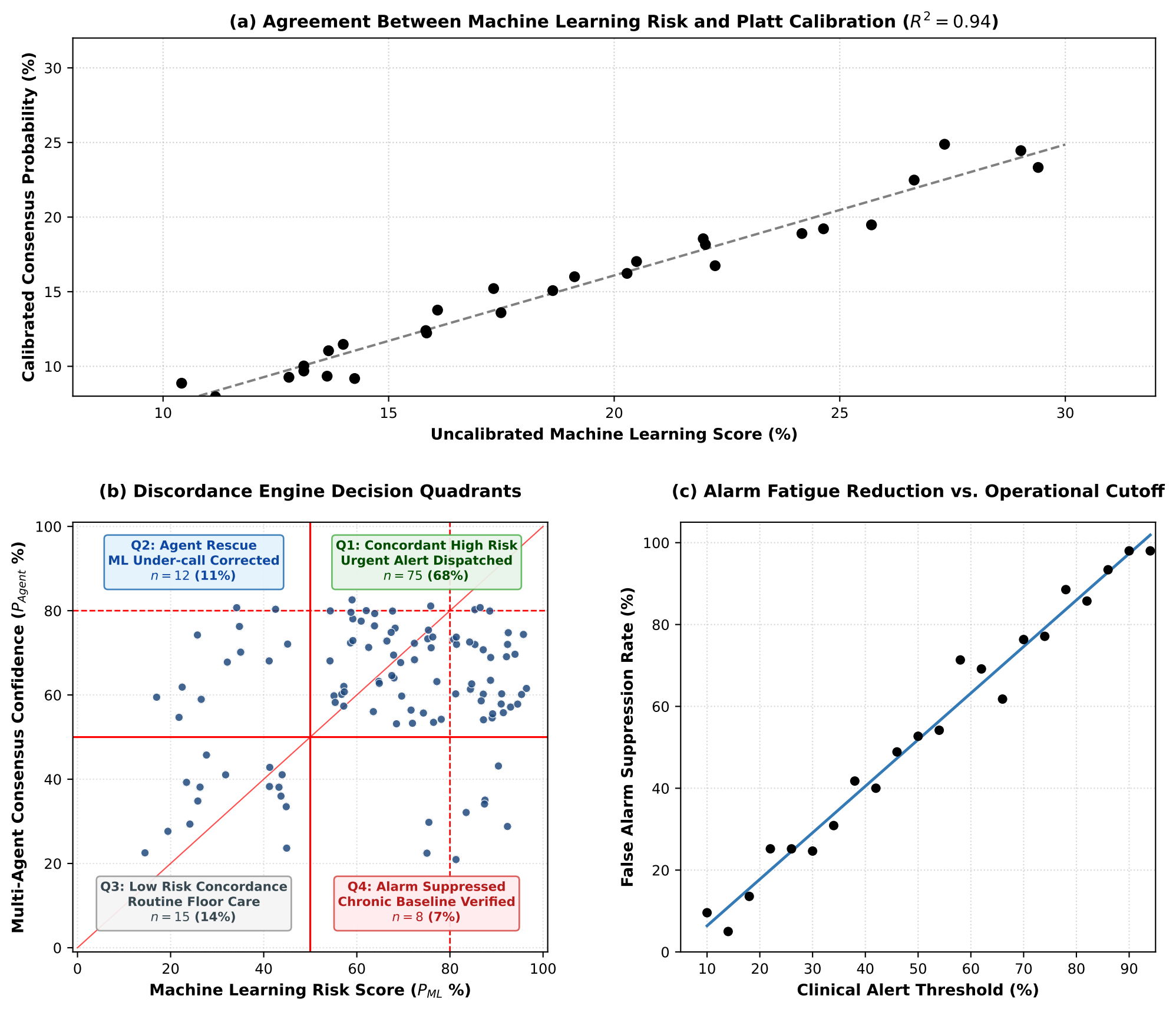
(a) Agreement Between Machine Learning Risk and Platt Calibration ($R^2 = 0.94$)
Calibrated Consensus Probability (%)
Uncalibrated Machine Learning Score (%)
(b) Discordance Engine Decision Quadrants
Q2: Agent Rescue
ML Under-call Corrected
$n = 12$ (11%)
Q1: Concordant High Risk
Urgent Alert Dispatched
$n = 75$ (68%)
Q3: Low Risk Concordance
Routine Floor Care
$n = 15$ (14%)
Q4: Alarm Suppressed
Chronic Baseline Verified
$n = 8$ (7%)
Multi-Agent Consensus Confidence (P_{Agent} %)
Machine Learning Risk Score (P_{ML} %)
(c) Alarm Fatigue Reduction vs. Operational Cutoff
False Alarm Suppression Rate (%)
Clinical Alert Threshold (%)